

Structure of a Culinary Ingredient Network and the SIS Phase Transition on Random Graphs

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ABSTRACT: This report studies complex networks from two complementary sides, structure and dynamics, linked by the configuration model. First we characterize the flavor network of Ahn *et al.*, where two ingredients are joined when they share a flavor compound, on its giant connected component. We compute its degree distribution and then compare its degree correlations, clustering and community structure with a degree-exact null model (double-edge-swap). The network is very dense and its degree distribution is broad but not scale-free. The degrees are strongly constraining: roughly 70% of the links are fixed by the degree sequence, so the null can only rearrange a small part of the graph. As a result the nearest-neighbor degree and the clustering are essentially reproduced by the null and are therefore largely explained by the degrees. The community structure, whose modularity ($Q \approx 0.12$) is about three times the null value ($Q \approx 0.045$), signals a weak but real modular organization. Second, we characterize the SIS non-equilibrium phase transition on configuration-model networks ($k_{\min} = 4$, $\gamma = 2.5$ and 3.5) with the lifespan method of Mata *et al.* [2] and an exact Gillespie algorithm. The peak-height exponents are shown for both γ . For $\gamma = 2.5$ the peak-shift fit resolves a small but non-zero $\lambda_c = 0.0026(10)$, consistent with the zero threshold expected for $\gamma \leq 3$, whereas for $\gamma = 3.5$ the fit is compatible with $\lambda_c = 0$ over our range of N , a limitation of the sizes rather than of the method.

I. INTRODUCTION

Complex networks can be studied from two sides: how the nodes are connected (structural), and by the dynamics that run on top of them. This report looks at both. We first characterize the structure of a real network and compare it with a null model. Then, we generate synthetic networks and use them to run a dynamical process.

The real network is the flavor network of Ahn *et al.* [1], which was originally built to look at the *food pairing hypothesis*: the idea that two ingredients go well together when they share many of the chemical compounds that give them their flavor. We use the network to measure its structural properties (degree distribution, degree correlations, clustering and community structure), and we compare them with a Configuration Model (CM) null ensemble.

For the second part we use the same configuration model, but now to generate scale-free networks on which we run a dynamical process, the susceptible–infected–susceptible (SIS) epidemic model. SIS dynamics have an absorbing phase with no infection and an endemic phase with a non-zero steady prevalence, and the transition between them is a non-equilibrium phase transition. The objective is to characterize it with the lifespan method [2], estimating the critical infection rate λ_c and the finite-size critical exponents.

All files, data and code can be openly accessed through my GitHub [repository](#).

II. DATA AND NETWORK REPRESENTATION

The flavor network of Ahn *et al.* [1] comes from a bipartite ingredient-compound network: nodes are ingredients and edges join two ingredients that share at least one compound. The original links are weighted by the number of shared compounds, but since our goal is to study the network’s topology, we will not consider it. We restrict the analysis to the giant connected component (GCC), to keep the network connected and disregard small node groupings (as only 11 nodes out of 1507 are not in the GCC).

The network is stored with an adjacency-array structure: a vector D of length N with the degrees, a vector V of length $2E$ with all the neighbour lists concatenated, and two pointers per node marking where its block starts in V and how much of it is filled. The file is read in two passes, the first to count degrees and the second to write neighbours.

As the network is not directed, the average degree comes from $\langle k \rangle = 2E/N$. In Table I are listed the basic properties of the network, which is very connected, as the average number of neighbours of a node is 296. The networks used in the second part are not empirical but synthetic configuration-model graphs, generated as explained in the methods.

Property	Value
Number of nodes N	1496
Number of edges E	221 777
Average degree $\langle k \rangle$	296
Maximum degree $k_{\max}$	990

TABLE I: Properties of the GCC of the flavor network.

III. METHODS

A. Structural descriptors

The degree distribution is the fraction of nodes with degree k , $P(k) = N_k/N$, and its complementary cumulative is $\text{ccP}(k) = \sum_{k' \geq k} P(k')$. The average nearest-neighbor degree is

$$\bar{k}_{nn}(k) = \frac{1}{N_k} \sum_{i \in v(k)} \frac{1}{k_i} \sum_j A_{ij} k_j, \quad (1)$$

with $v(k)$ the set of nodes of degree k and A_{ij} the adjacency matrix. We normalize it by $\kappa = \langle k^2 \rangle / \langle k \rangle$, the expected neighbor degree in an uncorrelated graph: a flat $\bar{k}_{nn}(k)/\kappa \approx 1$ means a neutral network, an increasing (decreasing) one means assortative (disassortative) mixing. The clustering coefficient by degree is

$$\bar{c}(k) = \frac{1}{k(k-1) N_k} \sum_{i \in Y(k)} 2T_i, \quad (2)$$

where T_i is the number of triangles at node i , found by checking, for every node, which pairs of its neighbors are also linked. The global clustering is $\bar{c} = \sum_k P(k) \bar{c}(k)$.

B. Community detection and configuration-model null

The communities are found with the Louvain algorithm, which maximizes the modularity

$$Q = \frac{1}{2E} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2E} \right) \delta(c_i, c_j), \quad (3)$$

where $\delta(c_i, c_j) = 1$ if i and j are in the same community. The algorithm starts with each node in its own community, moves nodes to neighbouring communities whenever Q grows, collapses each community into a single node, and repeats. It is fast and needs no input number of communities, but it is stochastic: the order in which nodes are visited can change the partition. We have done it using the *networkX* library in Python.

To see whether the measurements above follow from the degree sequence alone, we compare the network with a Configuration Model (CM) null ensemble. The CM gives each node k_i edges and pairs them at random, so every realization keeps the same degree sequence and the same $P(k)$, while destroying all other correlations. In practice, the Python function *configuration_model* creates self-loops and multiple edges. As we don't want those, we use the double-edge-swap method instead: starting from the real graph, we repeatedly pick two edges at random and swap their endpoints whenever the swap does not produce a self-loop or a repeated

edge. As our network is very dense, most attempted swaps are rejected, since the chosen endpoints are usually already connected, so the procedure is computationally heavy; we run it until each edge has been swapped several times and average $\bar{k}_{nn}(k)$, $\bar{c}(k)$ and the modularity over 100 independent realizations.

The same configuration-model idea appears in the second part of this work, but used the other way around: we do not randomize a real graph but impose a power-law degree sequence $P(k) \propto k^{-\gamma}$ and build the graphs from it, to obtain the synthetic scale-free networks on which the SIS dynamics runs.

C. SIS dynamics and the Gillespie algorithm

In the SIS model each node is either susceptible (S) or infected (I). An infected node recovers at rate $\delta = 1$, and infects each susceptible neighbour at rate λ , so λ is the only control parameter and time is measured in units of the mean recovery time. The state with no infected node is absorbing. We simulate the dynamics in continuous time with the Gillespie algorithm. At time t we keep the number of infected nodes $N_I(t)$ and the number of active edges $E_{\text{act}}(t)$ (edges with one infected and one susceptible node). Recoveries happen at total rate $\delta N_I \rightarrow N_I$ and infections at total rate λE_{act} , so the total event rate is

$$R(t) = N_I(t) + \lambda E_{\text{act}}(t). \quad (4)$$

The next event is then an infection with probability $\lambda E_{\text{act}}/R$. If this is not the case, it will be a recovery, and the time to the next event is drawn from $\Delta t = -\ln \xi / R$ with ξ uniform in $(0, 1)$.

It is important to keep the active-edge list updated without having to scan the whole network after every event. The graph is stored in CSR format, so the neighbours of any node can be read directly without going through the whole edge list. In addition, we keep two dynamic lists: one with the currently infected nodes and one with the active edges (which join an infected node to a susceptible one). Each list has an array that records, for every node or edge, the position it occupies in its list. This is what lets us insert and delete at constant time, swapping the element with the last one in the list. Then, when a node changes state (an infection or a recovery) we only need to reclassify the edges around it, at a cost of order $O(k)$, rather than reading all E edges. This way, we can work with networks up to $N = 10^6$ nodes.

D. Lifespan method and finite-size scaling

Each realization starts from a single infected node, chosen uniformly among the nodes of minimum degree $k_{\min} = 4$, and is declared *endemic* once the fraction of distinct nodes that have been infected (coverage) reaches $1/2$. If not, it is a finite outbreak with

a lifespan τ , which is the time until the infection dies out. The order parameter is the endemic probability $P_{\text{end}}(\lambda, N)$, and the susceptibility-like observable is the average lifespan of these outbreaks $\langle \tau(\lambda, N) \rangle$ (we also compute $\langle \tau^2 \rangle$). Near the transition they obey the finite-size scaling forms

$$P_{\text{end}}(\lambda, N) = N^{-\beta/\nu} f[(\lambda - \lambda_c)N^{1/\nu}], \quad (5)$$

$$\langle \tau^n(\lambda, N) \rangle = N^{\gamma_n/\nu} G_n[(\lambda - \lambda_c)N^{1/\nu}]. \quad (6)$$

From the second relation the peak of the lifespan curve drifts as $\lambda_p(N) = \lambda_c + aN^{-1/\nu}$, which we fit (Eq. 16 of [2]) to obtain λ_c and $1/\nu$. The peak position of each N is fitted with a local parabola through the maximum and its two neighbours, and the peak height then gives $\langle \tau^n \rangle_{\text{peak}} \sim N^{\gamma_n/\nu}$. Finally, the order parameter at the critical point gives $P_{\text{end}}(\lambda_c, N) \sim N^{-\beta/\nu}$.

We generate CM networks with $P(k) \propto k^{-\gamma}$, $k_{\text{min}} = 4$, $\gamma = 2.5$ and 3.5 , and $N = 10^4, 3 \times 10^4, 5 \times 10^4, 10^5, 3 \times 10^5, 5 \times 10^5, 10^6$. We use for k_{max} the largest expected degree in a power-law $k_{\text{max}} \sim N^{1/(\gamma-1)}$, and self-loops and multi-edges are removed. For every (γ, N) , we compute λ around the peak of $\langle \tau \rangle$ with 5×10^4 repetitions per point.

IV. RESULTS

A. Structure of the flavor network

The degree distribution (Fig. 1, left) is right-skewed: many ingredients have a relatively low degree, but the tail is very long and reaches $k_{\text{max}} \approx 990$. This comes from the density of the graph, not from a scale-free mechanism: a fit of the binned distribution gives an exponent close to 0.7, far from the 2–3 of scale-free networks.

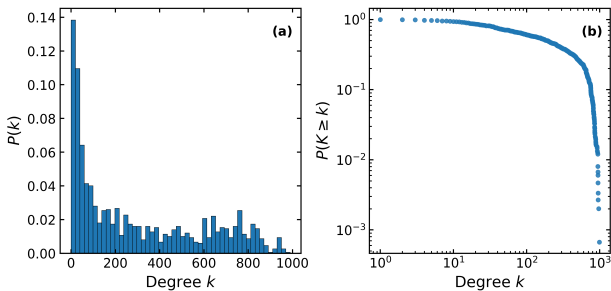


FIG. 1: **Degree distributions** (a): degree distribution $P(k)$ using 50 bins. (b): complementary cumulative distribution. Almost constant until reaching higher degree nodes.

The normalised average nearest-neighbor degree (Fig. 2, left) grows at small k and then decreases for higher values, showing an almost linear tendency

starting at middle k values. This behaviour expresses a disassortative tendency, where highly connected nodes preferentially connect with lower degree nodes. This indicates that ingredients sharing compounds with many others tend to be linked to ingredients that share fewer.

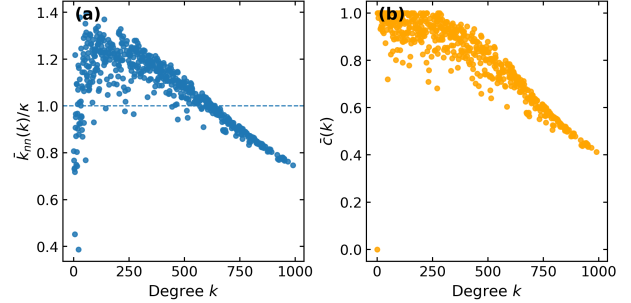


FIG. 2: **Nearest-neighbours and Clustering** (a): normalised average nearest-neighbours distribution. It increases for small degrees and falls for higher degrees. (b): clustering coefficient per degree. It falls with degree until reaching a clustering of ≈ 0.4

The clustering coefficient (Fig. 2, right) is close to one for low-degree nodes and drops slowly to ~ 0.4 for the most connected ones, with a global value $\bar{c} \approx 0.86$. Such a high clustering is expected in a dense graph, where two neighbours of a node are likely to be linked just by chance. The drop with k indicates that the ingredients sharing compounds with many others (higher degree) tend to act as bridges between more clustered local groups.

On the GCC, the Louvain algorithm finds six communities with $Q \approx 0.12$ (Fig. 3, left). Most nodes fall into two large groups and the rest into a few small ones. Some of these small groups are very homogeneous (for example one made almost only of conifer species and resins, another of exotic citrus oils), which is the kind of chemically similar cluster that Louvain can isolate for our ingredient network.

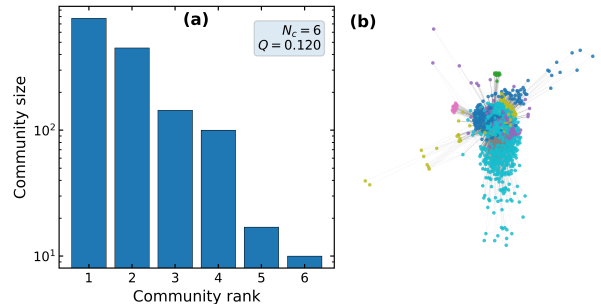


FIG. 3: **Community organization of the ingredient-pair network.** (a): ranked community size distribution obtained from Louvain community detection, displayed on a logarithmic scale. (b): network representation colored by community membership, showing the modular organization of the giant connected component.

A modularity of $Q = 0.12$ is low, but this is expected: in a dense graph where almost every node touches every group there is little room for a clean separation, so a low Q is normal. This matches [1], where the full network is too dense to be informative and the clear food classes (meats, fruits, dairy, etc.) only show up after keeping the statistically significant links (using weights). We use the full network, so what we measure is the weaker signal.

To see whether these measurements follow from the degree sequence alone, we compare them with the CM ensemble. Our network is so dense that the null double-edge-swap is strongly constrained: when we reshuffle the links keeping the degrees, only about 30% of them can actually move, so $\sim 70\%$ of every randomized graph is identical to the real one, because those edges must be present in any graph with this degree sequence. This is a result in itself, since the degrees alone fix most of the connections. However, it also limits the comparison for the local descriptors: because most of the graph we compare against is the real graph, the fact that the nearest-neighbor degree and the clustering barely differ from the null is a direct consequence of this, and should not be considered a strong test. Indeed, the real and null $\bar{k}_{nn}(k)$ nearly coincide (Fig. 4 (left)), and the real clustering ($\bar{c} \approx 0.86$) is only slightly above the null ($\bar{c} \approx 0.83$, Fig. 4 (right)). Finally, the disassortative shape of $\bar{k}_{nn}$ is the expected effect of having such large hubs, which are forced to connect to the low-degree majority.

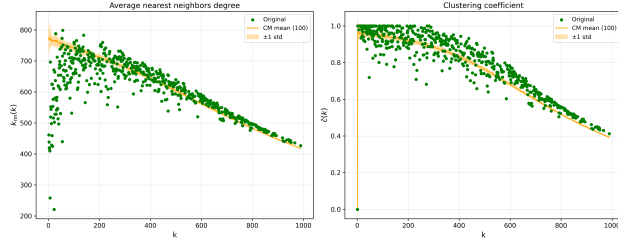


FIG. 4: **Comparison between the empirical ingredient-pair network and configuration model.** (left) average nearest-neighbour degree as a function of node degree. The empirical network is shown together with the mean behaviour obtained from configuration model realizations and the associated one-standard-deviation interval. (right) degree-dependent clustering coefficient for the empirical network and configuration model ensemble. Interval is associated with the one-standard-deviation interval.

We also ran Louvain on the random realizations. The modularity drops to $Q = 0.045$, against $Q \approx 0.12$ in the real network. Then, this comparison becomes more informative than the ones in Fig. 4, as modularity has a consistent drop and the small communities that were formed before dissolve. We can assume that the community structure that lives in the original network is part of the lower degree nodes, which are the ones reshuffled in our CM realisation.

B. SIS phase transition via the lifespan method

The average lifespan $\langle \tau(\lambda, N) \rangle$ (Fig. 5(a)) shows clear peaks that move towards smaller λ as N increases. The endemic probability $P_{\text{end}}(\lambda, N)$ (Fig. 5(b)) stays at zero and then rises sharply once $\lambda = \lambda_p(N)$ is large enough, with the onset shifting to smaller λ for larger networks. Both plots show the finite-size behaviour of the transition and locate the pseudo-critical point $\lambda_p(N)$, which drifts with N as $\lambda_p(N) = \lambda_c + aN^{-1/\nu}$. Fitting this shift (Fig. 5(c)) we get $\lambda_c = 0.0026 \pm 0.0010$ and $1/\nu = 0.369 \pm 0.040$ for $\gamma = 2.5$, and an almost zero λ_c with $1/\nu = 0.068 \pm 0.099$ for $\gamma = 3.5$ (Table II).

Exponent	$\gamma = 2.5$	$\gamma = 3.5$
λ_c	0.0026 ± 0.0010	~ 0
$1/\nu$	0.369 ± 0.040	0.068 ± 0.099
γ_1/ν	-0.03	-0.002
γ_2/ν	0.59	0.42
β/ν	0.52	0.75

TABLE II: Estimated exponents from the lifespan method (peak-shift fit, peak-height scaling and $P_{\text{end}}(\lambda_p(N), N)$).

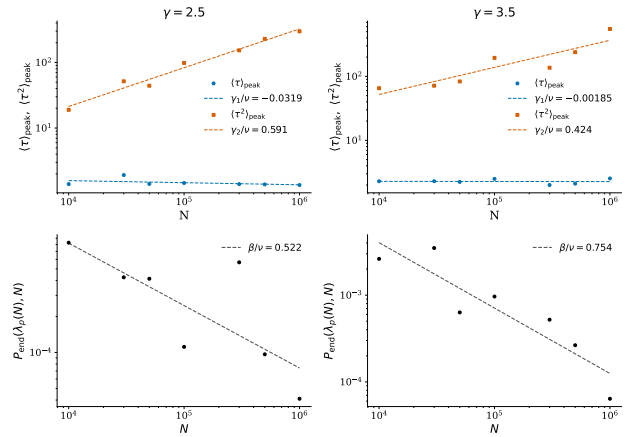


FIG. 6: **Top:** peak-height scaling $\langle \tau \rangle_{\text{peak}}, \langle \tau^2 \rangle_{\text{peak}} \sim N^{\gamma_n/\nu}$. **Bottom:** order parameter at the peak, $P_{\text{end}}(\lambda_p(N), N) \sim N^{-\beta/\nu}$. $\gamma = 2.5$ (left), $\gamma = 3.5$ (right).

The height of the $\langle \tau \rangle$ peak barely changes with N (Fig. 6, top), so γ_1/ν is compatible with zero for both γ . This is a direct consequence of $\delta \simeq 1$ of the lifespan distribution, in which $\langle \tau \rangle$ diverges logarithmically at the critical point and cannot resolve a clean value for the exponent [2]. The second moment is free of this, and $\langle \tau^2 \rangle_{\text{peak}}$ grows as a power law and gives a non-zero and positive γ_2/ν for both γ (Table II). The order parameter P_{end} at λ_p (Fig. 6, bottom) also decreases as $N^{-\beta/\nu}$, which fixes β/ν .

In Fig. 7 we rescale the curves with the exponents of Table II and plot them against $(\lambda - \lambda_c)N^{1/\nu}$. For $\gamma = 2.5$ (top row) the $\langle \tau \rangle$ and P_{end} curves collapse reasonably onto a single curve, despite a considerable

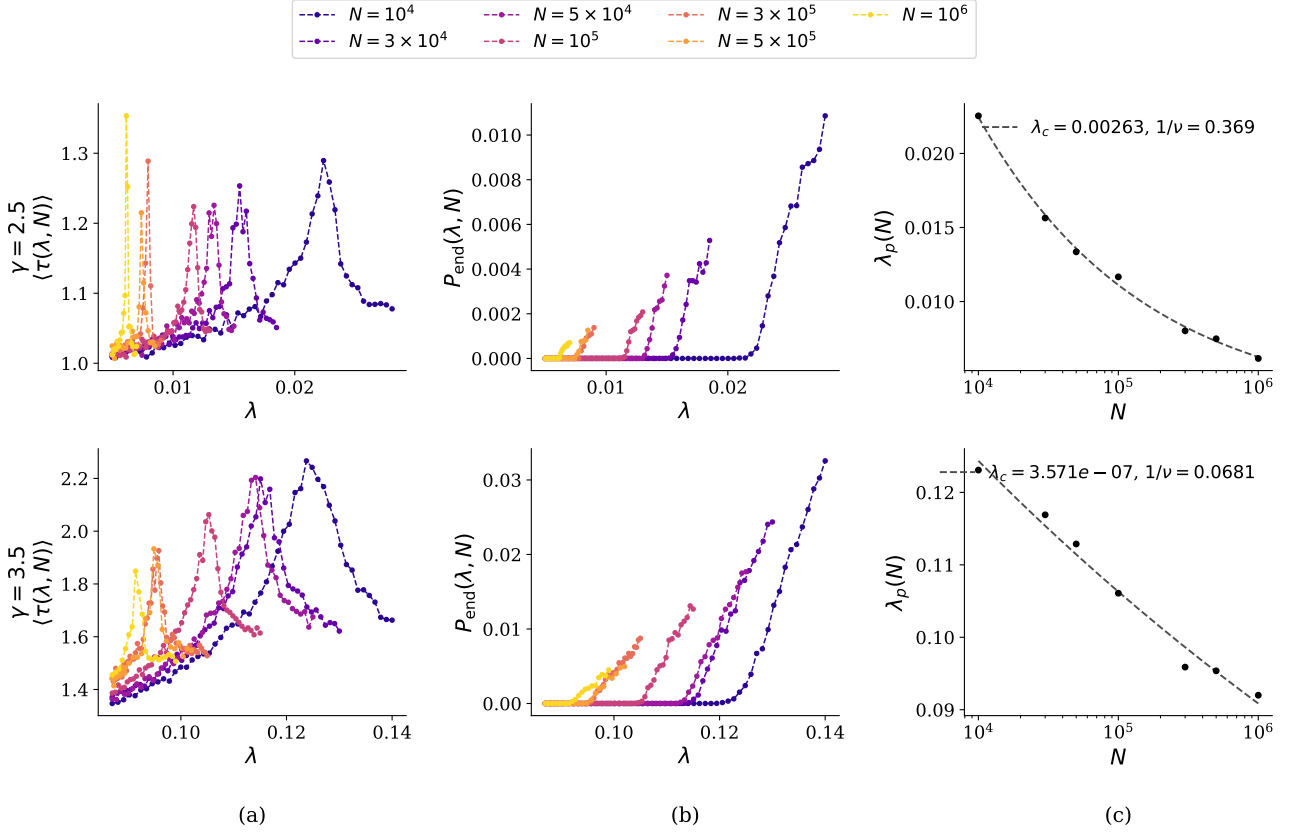


FIG. 5: **Lifespan-method curves** for $\gamma = 2.5$ (top row) and $\gamma = 3.5$ (bottom row), with one colour per size N . **(a)** average lifespan $\langle \tau(\lambda, N) \rangle$ vs. λ ; **(b)** endemic probability $P_{\text{end}}(\lambda, N)$ vs. λ ; **(c)** peak position $\lambda_p(N)$ vs. N with the peak-shift fit $\lambda_p(N) = \lambda_c + aN^{-1/\nu}$ (Eq. 16 of [2]), which gives λ_c and $1/\nu$.

deviation for the $N = 10^6$ peak of $\langle \tau^2 \rangle$, which is bigger than the rest. For $\gamma = 3.5$ (bottom row) the $\langle \tau^2 \rangle$ curves also overlap, but this is much weaker, since $1/\nu \approx 0.07$, the factor $N^{1/\nu}$ barely changes across our sizes (about 1.4 between $N = 10^4$ and 10^6 , against 5.5 for $\gamma = 2.5$), so the curves are hardly stretched despite their overlapping. Consistently, the P_{end} collapse is tighter for $\gamma = 2.5$ than for $\gamma = 3.5$, where the curves rise at slightly different points.

For $\gamma = 2.5$ the peak positions $\lambda_p(N)$ flatten through the whole range of N and tend to a small but non-zero value (Fig. 5(c), top row), so the fit is able to compute $\lambda_c = 0.0026 \pm 0.0010$ together with a well defined $1/\nu = 0.369 \pm 0.040$, and the collapse is clean (Fig. 7, top row). This is what the theory predicts: for $2 < \gamma \leq 3$ the epidemic threshold vanishes as $N \rightarrow \infty$ because $\langle k^2 \rangle$ diverges with the size [3], so a small λ_c decreasing at finite N is the expected behaviour. For $\gamma = 3.5$, instead, $\lambda_p(N)$ is compatible with a pure power law ($\lambda_c = 0$) over the whole range: the fit cannot separate λ_c from zero and $1/\nu$ is smaller than its uncertainty ($1/\nu = 0.068 \pm 0.099$), so the two become degenerate and the collapse is worse (Fig. 7, bottom row). Here $\gamma > 3$ means the threshold should converge to a finite value, but we cannot see this convergence (with $N \leq 10^6$) as it is governed by the small

exponent $1/\nu \approx 0.068$.

V. CONCLUSIONS

We have described the flavor network with a set of structural measurements and compared them with a degree-exact null model. The network is very dense and its degree distribution is broad but not scale-free, coming from the density. The comparison with the CM realisations, however, must be read with care: the network is so constrained that the null can only rearrange about 30% of the links, while 70% are fixed by the degrees. Most of the connections are degree-determined, and the null has little power to test the local descriptors. We find that the flavor network presents a disassortative $\bar{k}_{nn}$ and strong clustering but, as far as we can tell, these properties are what these degrees must produce. The property which the comparison can genuinely test, because it depends on the rearrangeable links, is the community structure, and there the real network exceeds the null. The conclusion is therefore that a weak but real modular organization is present beyond the degrees, which agrees with the literature [1].

For the dynamics, the lifespan method applied to SIS on CM networks gives robust peak-height exponents

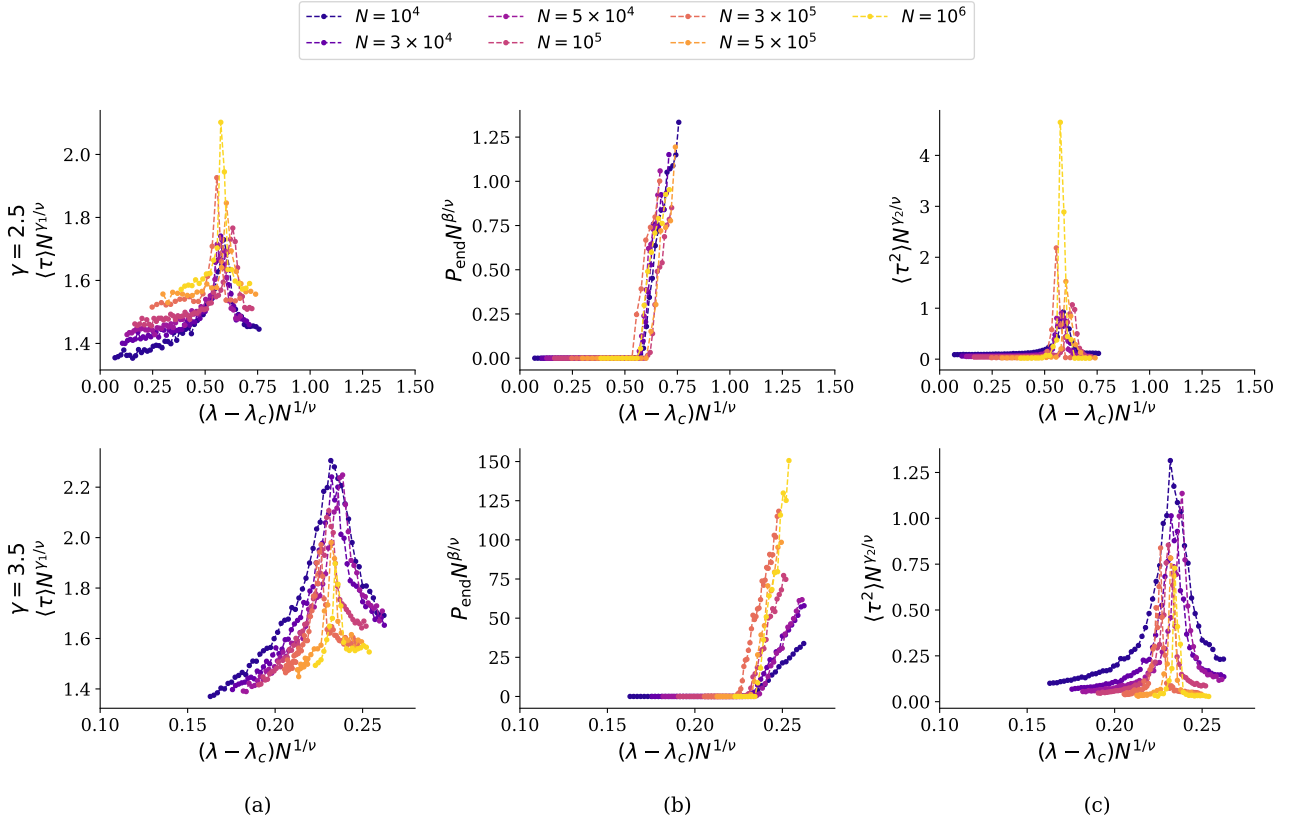


FIG. 7: **Finite-size collapse** for $\gamma = 2.5$ (top row) and $\gamma = 3.5$ (bottom row), plotted against $(\lambda - \lambda_c)N^{1/\nu}$. (a) $\langle \tau \rangle N^{\gamma_1/\nu}$; (b) $P_{\text{end}} N^{\beta/\nu}$; (c) $\langle \tau^2 \rangle N^{\gamma_2/\nu}$. All use λ_c and $1/\nu$ from Table II.

for both $\gamma = 2.5$ and 3.5 (with the expected $\gamma_1/\nu \approx 0$ of the first moment), while λ_c and $1/\nu$ are only reliably resolved for $\gamma = 2.5$, which agree with the theoretical

expectation of a 0 threshold for $\gamma \leq 3$, and the much slower finite-size convergence that affects $\gamma > 3$.

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 [2] A. S. Mata, M. Boguñá, C. Castellano and R. Pastor-Satorras, *Lifespan method as a tool to study criticality in absorbing-state phase transitions*, Phys. Rev. E **91**,

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